

Paper Practice 9.5**Geometric Sequences**

Show all of your work in the space provided. Use $a_n = a_1 \cdot r^{(n-1)}$ where a formula is needed. Round money to the nearest whole unit.

- 1.** Determine whether each sequence is geometric. If it is, state the common ratio. (Is It Geometric?)

a. $-432, 144, -48, 16, \dots$

b. $4, 9, 25, 36, \dots$

c. $16, 12, 8, 4, \dots$

- 2.** Find the next three terms in each geometric sequence. (Next Terms)

a. $64, 16, 4, 1, \dots$

b. $8, 12, 18, 27, \dots$

c. $3, -6, 12, -24, \dots$

- 3.** Find the next three terms in each geometric sequence. (Next Terms)

a. $96, 48, 24, 12, \dots$

b. $2, 10, 50, 250, \dots$

- 4.** Write an explicit formula $a_n = a_1 \cdot r^{(n-1)}$ for each sequence. (Explicit Formula)

a. $5, 15, 45, 135, \dots$

b. $800, 400, 200, 100, \dots$

- 5.** A sequence has $a_1 = 6$ and $r = -2$. Write the first five terms. (Build a Sequence)

- 6.** Use an explicit formula to find the indicated term. (Find a Term)

a. 512, 256, 128, 64, ... find a_{11}

b. -8, 24, -72, 216, ... find a_9

- 7.** Use an explicit formula to find the indicated term. (Find a Term)

a. 5, 3, 1.8, 1.08, ... find a_8

b. 7, 21, 63, ... find a_6

- 8.** North Dakota's population was 685,242 in 2011 and has been increasing about 2.5% each year. Estimate the population in 2030. (Apply)

- 9.** Global vinyl record sales were \$267 million in 2014 and have been increasing about 26% each year. Estimate sales for 2025. (Apply)

- 10.** Error Analysis – Gus says 2, 4, 6, 8, ... is geometric because the terms keep increasing. Identify his mistake. (Error Analysis)

- 11.** Explain the difference between an arithmetic and a geometric sequence, and give an example of each. (Explain)